Mohamed Moustafa Dawoud

PhD Student, Computer Science & Engineering
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AI Safety and Privacy
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RESEARCH INTERESTS

AI safety and privacy; auditing deployed systems against the rules that govern them; AI-enabled harm and its commercialization; privacy regulation and enforcement; machine learning and large language models as measurement instruments; safety and privacy of AI agents across contexts and jurisdictions.

EDUCATION

University of California, Santa Cruz 2025 – present
Ph.D. in Computer Science & Engineering
Co-advised by Prof. Ram Sundara Raman and Prof. Leilani Gilpin. GPA 3.95/4.00. Expected June 2029.

University of California, Santa Cruz 2025 – 2026
M.S. in Computer Science & Engineering
Committee: Prof. Ram Sundara Raman, Prof. Álvaro Cárdenas, Prof. David T. Lee, Dr. Alejandro Cuevas.

German International University, Cairo, Egypt 2021 – 2024
B.Sc. in Informatics & Computer Science, Major in IT Security
Excellent with High Honors (GPA 1.08, German scale; 0.7 highest). Full merit scholarship.

PUBLICATIONS

Author names appear in publication order; my name is in bold. † denotes first authorship.

Governing the Open-Weight Ecosystem Behind Explicit Image Generation†
Mohamed Moustafa Dawoud (lead author)
In preparation. Measurement of the open-weight model ecosystem behind explicit and identity-specific image generation, including non-consensual intimate imagery (NCII): model-level interventions, their effect on shared-model use, and cross-platform persistence, to inform platform governance. 2026

The Impact of US Age Verification Laws on Adult Content Consumption
Alejandro Cuevas, **Mohamed Moustafa Dawoud**, Zhibin Shen, Ram Sundara Raman, and Manoel Horta Ribeiro
Under review. Individual-level panel data from mobile browsing activity, used to estimate the effect of state age-verification laws on adult-content consumption at compliant and non-compliant sites. 2026

PrivAudit: A Dual-Lens Auditing Framework for Website Privacy Practices under the CCPA†
Mohamed Moustafa Dawoud, Riya Aggarwal, Likith Rahul Krishnamurthy, and Ram Sundara Raman
33rd ACM SIGSAC Conference on Computer and Communications Security (CCS). To appear. 2026

From Underground to Mainstream Marketplaces: Measuring AI-Enabled NSFW Deepfakes on Fiverr†
Mohamed Moustafa Dawoud, Alejandro Cuevas, and Ram Sundara Raman
Symposium on Usable Security and Privacy (USEC), co-located with NDSS. Presented in San Diego, CA 2026
| *[Video]*

Vendor communication themes in darknet Ransomware-as-a-Service (RaaS) advertisements
Taylor Fisher, Zacharias Pieri, C. Jordan Howell, Roberta O'Malley, Lauren Tremblay, and **Mohamed Dawoud**
Computers in Human Behavior, 165:108571. 2025

DVa: Extracting Victims and Abuse Vectors from Android Accessibility Malware
Haichuan Xu, Mingxuan Yao, Runze Zhang, **Mohamed Moustafa Dawoud**, Jeman Park, and Brendan Saltaformaggio
33rd USENIX Security Symposium (USENIX Security 24). 2024

Social Engineering and Technical Security Fusion

Wassim Alexan, Eyad Mamdouh, Mohamed ElBeltagy, Ahmed Ashraf, **Mohamed Moustafa**, and Hashem Al-Qurashi
International Telecommunications Conference (ITC-Egypt). 2022

Image Encryption Through Rössler System, PRNG S-Box and Recamán's Sequence
Mohamed ElBeltagy, Wassim Alexan, Abdelrahman Elkhamry, **Mohamed Moustafa**, and Hisham H. Hussein
IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC). 2022

IoMT Security: SHA3-512, AES-256, RSA and LSB Steganography
Wassim Alexan, Ahmed Ashraf, Eyad Mamdouh, Sarah Mohamed, and **Mohamed Moustafa**
8th NAFOSTED Conference on Information and Computer Science (NICS). 2021

SOFTWARE

PrivAudit 2026
Open-source dual-lens auditing framework pairing LLM-based analysis of privacy-policy disclosures with browser-based measurement of cookie tracking, to audit CCPA compliance at scale.

PaperClerk.ai 2026
Privacy-preserving research-paper triage tool running entirely on-device with local LLMs. Python, Apache-2.0.

RESEARCH EXPERIENCE

California Privacy Protection Agency (CPPA), via University Enterprises, Inc. Jun 2026 – present
Research Technologist Intern, Audits Division

- Reporting to Chief Privacy Auditor Sabrina Ross and Dr. Christina Yeung at the first U.S. agency established solely to enforce data privacy law.
- Support the Agency's audits of businesses' data-handling, security, and consumer-privacy practices under the California Consumer Privacy Act.
- Evaluate privacy and governance risks in automated decision-making and emerging AI systems to inform enforcement and policy priorities.

University of California, Santa Cruz Jun 2025 – present
Graduate Student Researcher, RANDLab

- Research on AI safety and privacy, auditing whether deployed systems behave as their policies, labels, and regulations claim. First-author work at CCS 2026 and USEC 2026.
- Collaborate with Stanford on a practitioner interview study of how privacy and AI governance frameworks are interpreted in practice.

Mohamed bin Zayed University of Artificial Intelligence Sep – Dec 2024
Research Associate

- Research on combating deepfakes and promoting responsible use of AI through a human-centered lens.

Georgia Institute of Technology Oct 2023 – Feb 2024
Research Intern

- Conducted systems and network security research on Android malware analysis under Prof. Brendan D. Saltaformaggio.
- Contributed to **DVa** (USENIX Security 2024), which extracts targeted victims and abuse vectors from Android accessibility malware through dynamic, victim-guided execution and abuse-vector-guided symbolic analysis.
- Led (first author) a separate study on using large language models to assist static analysis of Android malware: designed and executed a large-scale experiment on a manually curated dataset of benign and malicious Android code collected in the wild, demonstrating strong LLM capabilities for reasoning about malicious behaviors and summarizing complex code patterns.

University of South Florida Jan – Aug 2023
Undergraduate Research Assistant (Remote)

- Cybercrime and underground economies with Prof. C. Jordan Howell, focused on Ransomware-as-a-Service marketplaces.
- Led the threat-intelligence component: collection and analysis of darknet RaaS vendor advertisements.

German University in Cairo Sep 2021 – Jan 2023
Information Security Research Intern

- Applied security research with Prof. Wassim Alexan on cryptography and IoT/IoMT security, resulting in three publications on image encryption and secure medical data transmission.

TEACHING EXPERIENCE

University of California, Santa Cruz Mar – Jun 2026
Graduate Teaching Assistant — CSE 115C: Software Design Project II

- Capstone software engineering course. Supported student teams through design and implementation, running weekly check-ins, code reviews, and design feedback. Supervisor: Richard Jullig.

Dartmouth College Jan – Jun 2025
Lead Graduate Teaching Assistant — COSC 55: Security & Privacy

- Lead TA for an advanced undergraduate course on computer and network security (28 students) with Prof. Sami Saydjari.
- Designed an AI-integrated teaching model that treated tools such as ChatGPT and GitHub Copilot as occasions for explanation and reflection rather than rote code production.
- Created a new lab series emphasizing design freedom and oral justification of design choices; ran all grading, oral evaluations, and rubric design.

HONORS & AWARDS

AAAS CASE Workshop Delegate, Dartmouth College 2025
One of four graduate students selected to represent Dartmouth at the AAAS *Catalyzing Advocacy in Science and Engineering* Workshop; engaged with federal policymakers in Washington, D.C.

Full Merit Scholarship, German International University 2021
Awarded to top-ranked national students.

18th Nationwide, Thanawya Amma Mathematics 2021
Ranked 18th nationally among ∼600,000 students; 408/410 (99.51%).

SERVICE & OUTREACH

Sub-reviewer, ACM Internet Measurement Conference (IMC) 2026
Reviewed three submissions at the invitation of a program committee member.

Co-founder & Lead Maintainer, Egyptians in CS Research 2026
Community directory of 262 Egyptian computer science researchers across 16 research tracks. Expanded from AI to all of computer science with the original creator, Badr AlKhamissi.

PhD Application Mentorship Ongoing
Mentor a cohort of prospective PhD applicants each cycle, with an emphasis on students who have had limited access to research mentorship. Feedback on statements of purpose, research statements, and CVs; advice on program selection and strategy.

Dartmouth Guarini School Student Spotlight 2025
Profiled on research in human-centered security and privacy, the path from Zefta, Egypt to Dartmouth, and advocacy for international students in STEM policy.

Conference Volunteer, IEEE Secure Development Conference (SecDev) 2023
Session logistics and attendee coordination. Atlanta, GA.